

**Time:** 5 minutes**Outcome:** Analyze Profit-Maximizing Output & the $MR = MC$ Rule.**Difficulty:** ●●○ Intermediate

Profit-Maximizing Output & the $MR = MC$ Rule



THE CORE IDEA

For a price-taking firm, marginal revenue (MR) equals market price. An additional unit raises profit when MR exceeds marginal cost (MC) and lowers profit when MC exceeds MR. The operating optimum is where rising MC crosses MR, provided producing covers avoidable costs.



HOW TO RECOGNIZE IT

- Use the firm's marginal-revenue curve and the rising portion of marginal cost.
- Choose the output where marginal cost equals marginal revenue, provided the firm should operate.
- If marginal revenue exceeds marginal cost, another unit raises profit; if marginal cost exceeds marginal revenue, it lowers profit.
- Do not choose output by minimizing average total cost unless that output also satisfies the marginal rule.

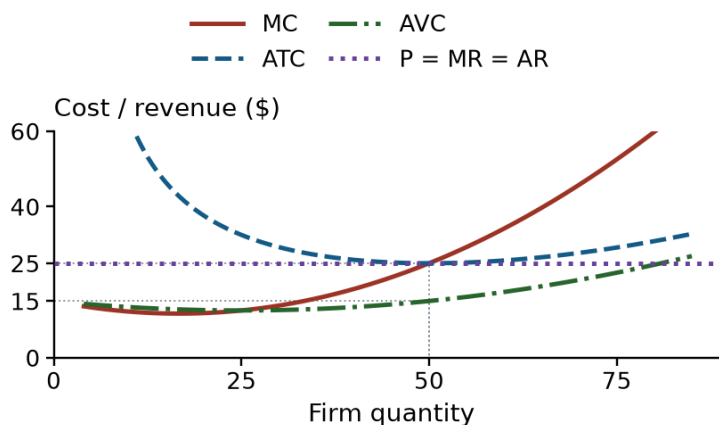


WATCH OUT

The marginal crossing identifies the best positive output. Compare revenue with avoidable costs before choosing that output over shutdown.



WORKED EXAMPLE: USING THE $MR = MC$ RULE



The firm's market price and marginal revenue are \$25. The rising marginal-cost curve reaches \$25 at 50 units, so the last unit adds exactly as much revenue as cost. Producing fewer units leaves profitable output unmade; producing more makes MC exceed MR. Profit-maximizing output is 50.



CHECK YOURSELF

MR is \$25 and the next unit's MC is \$19. What should the firm do?

**READY?**

Return to the game and master this concept.